Lecture - 14

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

TESTING OF HYPOTHESIS FOR TWO POPULATION VARIANCES

we have already mentioned that before applying t-test for difference of two population means, one of the requirements is to check the equality of variances of two populations. This assumption can be checked with the help of F-test for two population variances. This F-test is also important in a number of contexts. For example, an economist may want to test whether the variability in incomes differ in two populations, a quality controller may want to test whether the quality of the product is changing over time, etc.

Assumptions

The assumptions for F-test for testing the variances of two populations are:

1. The populations from which the samples are drawn must be normally distributed.
2. The samples must be independent of each other.

Now, we come to the general procedure of this test.

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n_1 from a normal population with mean μ_1 and variance σ_1^2 . Similarly, $Y_1, Y_2, \ldots, Y_n$ be a random sample of size n_2 from another normal population with mean μ_2 and variance σ_2^2. Here, we want to test the hypothesis about the two population variances so we can take our alternative null and hypotheses as

$H_0: \sigma_1^2 = \sigma_2^2 = \sigma^2 \;\; and \; H_1: \sigma_1^2 \neq \sigma_2^2$ [for two-tailed test]

Or $\left.\begin{array}{c} H_0: \sigma_1^2 \leq \sigma_2^2 \;\; and \; H_1: \sigma_1^2 > \sigma_2^2 \\ H_0: \sigma_1^2 \geq \sigma_2^2 \;\; and \; H_1: \sigma_1^2 < \sigma_2^2 \end{array}\right\}$ [for one-tailed test]

For testing the null hypothesis, the test statistic F is given by

$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$ under H_0

$$S_1^2 = \frac{1}{n_1 - 1} \sum (X - \bar{X})^2, and \; S_2^2 = \frac{1}{n_2 - 1} \sum (Y - \bar{Y})^2$$

For computational simplicity, we can also write

$$S_1^2 = \frac{1}{n_1 - 1} \left(\sum X^2 - \frac{(\sum X)^2}{n_1} \right), and \; S_2^2 = \frac{1}{n_2 - 1} \left(\sum Y^2 - \frac{(\sum Y)^2}{n_2} \right),$$

The test statistic F follows F-distribution with $\nu_1 = n_1 - 1$, and $\nu_2 = n_2 - 2$ degrees of freedom as discussed.

After substituting the values of S_1^2 and S_2^2, we get calculated value of test statistic. Let F_{cal} be calculated value of test statistic F.

Obtain the critical value(s) or cut-off value(s) in the sampling distribution of the test statistic F and construct rejection (critical) region of size α. The critical values of the test statistic F for various df and different level of significance α are given in Table IV (F-table) of the Appendix at the end of Block 1of this course.

After doing all calculations discussed above, we have to take the decision about rejection or non-rejection of the null hypothesis. This is explained below:

In case of one-tailed test:

Case I: When $H_0 : \sigma_1^2 \leq \sigma_2^2 \ and \ H : \sigma_1^2 > \sigma_2^2$ (right-tailed test)

In this case, the rejection (critical) region falls at the right side of the probability curve of the sampling distribution of test statistic F.

Suppose $F_{(\nu_1,\nu_2),\alpha}$ is the critical value of test statistic F with ($\nu_1 = n_1 - 1$, $\nu_2 = n_2 - 1$) df at α level of significance so entire region greater than or equal to $F_{(\nu_1,\nu_2),\alpha}$ is the rejection (critical) region and less than $F_{(\nu_1,\nu_2),\alpha}$ is the non-rejection region.

If $F_{cal} \geq F_{(\nu_1,\nu_2),\alpha}$, that means calculated value of test statistic lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that samples data provide us sufficient evidence against the null hypothesis and there is Chi-square and F-Tests a significant difference between population variances.

If $F_{cal} < F_{(\nu_1,\nu_2),\alpha}$, that means calculated value of test statistic lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance. Therefore, we conclude that the samples data fail to provide us sufficient evidence against the null hypothesis and the difference between population variances due to fluctuation of sample.

Case II: When $H_0 : \sigma_1^2 \geq \sigma_2^2$ *and* $H : \sigma_1^2 < \sigma_2^2$ (left-tailed test)

In this case, the rejection (critical) region falls at the left side of the probability curve of the sampling distribution of test statistic F.

Suppose $F_{(\nu_1,\nu_2),(1-\alpha)}$, is the critical value at α level of significance then entire region less than or equal to $F_{(\nu_1,\nu_2),(1-\alpha)}$ is the rejection(critical) region and greater than $F_{(\nu_1,\nu_2),(1-\alpha)}$ is the non-rejection region.

If $F_{cal} \leq F_{(\nu_1,\nu_2),(1-\alpha)}$ that means calculated value of test statistic lies in rejection (critical) region, then we reject the null hypothesis H_0 at α level of significance.

If $F_{cal} > F_{(\nu_1,\nu_2),(1-\alpha)}$ that means calculated value of test statistic lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Note: F-table mentioned above gives only the right tailed critical values with different degrees of freedom at different level of significance. The left tailed critical value of F-test can always be obtained by the formula given below:

$$F_{(\nu_1,\nu_2),(1-\alpha)} = \frac{1}{F_{(\nu_2,\nu_1),(\alpha)}}$$

In case of two-tailed test:

When $H_0 : \sigma_1^2 = \sigma_2^2 \; and \; H : \sigma_1^2 \neq \sigma_2^2$

In this case, the rejection (critical) region falls at both sides of the probability curve of the sampling distribution of test statistic F and half the area(α) i.e., α/2 of rejection (critical) region lies at left tail and other half on the right tail.

Suppose $F_{(\nu_1,\nu_2),(1-\alpha/2)}$ and $F_{(\nu_1,\nu_2),(\alpha/2)}$ are the two critical values at the left-tailed and right-tailed respectively on pre-fixed α level of significance. Therefore, entire region less than or equal to $F_{(\nu_1,\nu_2),(1-\alpha/2)}$ and greater than or equal to $F_{(\nu_1,\nu_2),(\alpha/2)}$ are the rejection (critical) regions and between $F_{(\nu_1,\nu_2),(1-\alpha/2)}$ and $F_{(\nu_1,\nu_2),(\alpha/2)}$ is the non-rejection region.

If $F_{cal} \geq F_{(\nu_1,\nu_2),(\alpha/2)}$ or $F_{cal} \leq F_{(\nu_1,\nu_2),(1-\alpha/2)}$ that means calculated value of test statistic lies in rejection(critical) region, then we reject the null hypothesis H_0 at α level of significance.

If $F_{(\nu_1,\nu_2),(1-\alpha/2)} < F_{cal} < F_{(\nu_1,\nu_2),(\alpha/2)}$ that means calculated value of test statistic F lies in non-rejection region, then we do not reject the null hypothesis H_0 at α level of significance.

Procedure of taking the decision about the null hypothesis on the basis of p-value:

To take the decision about the null hypothesis on the basis of p-value, the p-value is compared with level of significance (α) and if p-value is less than or equal to α then we reject the null hypothesis and if the p-value is greater than α we do not reject the null hypothesis.

For F-test, p-value can be defined as:

For one-tailed test:

For $H_1 : \sigma_1^2 > \sigma_2^2$ (right-tailed test)

p-value = $P[F \geq F_{cal}]$

For $H_1 : \sigma_1^2 < \sigma_2^2$ (left-tailed test)

p-value = $P[F \leq F_{cal}]$

For two-tailed test: $H_1 : \sigma_1^2 \neq \sigma_2^2$

For two-tailed test the p-value is approximated as

p-value = 2P [F ≥ Fcal]

The p-value for F-test can be obtained with the help of F-table. Similar to t-test, this table gives F values corresponding to the standard values of α such as 0.10, 0.05, 0.025 and 0.01 only. Therefore, with the help of F-table, we cannot find the exact p-value. So, we can approximate p-value for this test.

For example, if test is right-tailed and calculated (observed) value of test statistic F is 2.65 with 24 degrees of freedom of numerator and 14 degrees of freedom of the denominator then we can find the p-value as: Since test statistic is based on (24, 14) df therefore, we move across all the values of tabulated F corresponding to (24, 14) df at α such as 0.10, 0.05, 0.025 and 0.01 and find the values in which calculated F-value falls. Since we have F tabulated with (24, 14) df at α = 0.10, 0.05, 0.025 and 0.01 as

α =	0.10	0.05	0.025	0.01
$F_{(24, 14), \alpha}$	1.94	2.35	2.79	3.43

Since calculated F-value (= 2.65) falls between 2.35 and 2.79, corresponding to one-tailed area α = 0.05 and 0.025 respectively therefore p-value lies between 0.0025 and 0.05, that is,

$$0.025 < \text{p-value} < 0.05$$

If in the above example the test is two-tailed then the two values 0.025 and 0.05 would be doubled for p-value, that is,

$$0.05 < \text{p-value} < 0.10$$

Let us do some examples based on this test.

Example 1: The following data relate to the number of items produced in a Chi-square and F-Tests shift by two workers A and B for some days:

A	26	37	40	35	30	30	40	26	30	35	45
B	19	22	24	27	24	18	20	19	25		

Assuming that the parent populations are normal, can it be inferred that B is more stable (or consistent) worker compared to A?

Solution: Here, we want to test that worker B is more stable than worker A.

As we know that stability of data is related to variance of the data. Smaller value of the variance implies data that it is more stable. Therefore, to compare stability of two workers, it is enough to compare their variances. If σ_1^2 *and* σ_2^2 denote the variances of worker A and worker B respectively then our claim is $\sigma_1^2 > \sigma_2^2$ and its complement is $\sigma_1^2 \leq \sigma_2^2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$H_0 : \sigma_1^2 \leq \sigma_2^2$

$H_1 : \sigma_1^2 > \sigma_2^2$ [worker B is more stable than worker A]

Since the alternative hypothesis is right-tailed so the test is right-tailed test.

Here, we want to test the hypothesis about two population variances and sample sizes $n_1 = 11 (< 30)$ and $n_2 = 9 (< 30)$ are small. Also populations under study are normal and both samples are independent so we can go for F-test for two population variances.

For testing the null hypothesis, test statistic is given by

$$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1} \tag{1}$$

Where, $S_1^2 = \frac{1}{n_1-1}\left(\sum X^2 - \frac{(\sum X)^2}{n_1}\right)$, *and* $S_2^2 = \frac{1}{n_2-1}\left(\sum Y^2 - \frac{(\sum Y)^2}{n_2}\right)$, Calculation for S_1^2 and S_2^2:

Items Produced by A (Variable X)	$(X-\bar{X})=(X-34)$	$(X-\bar{X})^2$	Items Produced by B (Variable Y)	$(Y-\bar{Y})=(Y-34)$	$(Y-\bar{Y})^2$
26	-8	64	19	-3	9
37	3	9	22	0	0
40	6	36	24	2	4
35	1	1	27	5	25
30	-4	16	24	2	4
30	-4	16	18	-4	16
40	6	36	20	-2	4
26	-8	64	19	-3	9
30	-4	16	25	3	9
35	1	1			
45	11	121			
Total: 374	0	380	198	0	80

$$\bar{X}=\frac{\sum X}{n_1}=\frac{374}{11}=34 \text{ and } \bar{Y}=\frac{\sum Y}{n_2}=\frac{198}{9}=22$$

$$, S_1^2=\frac{1}{n_1-1}\sum(X-\bar{X})^2=\frac{380}{10}=38$$

$$and\ S_2^2=\frac{1}{n_2-1}\sum(Y-\bar{Y})^2=\frac{80}{8}=10$$

Putting the value of S_1^2 and S_2^2: in equation (1), we have

$$F=\frac{38}{10}=3.8$$

The critical (tabulated) value of test statistic F for right-tailed test corresponding

(n_1-1, n_2 -1) = (10, 8) df at 1% level of significance is

$$F_{(n_1-1,n_2-1),\alpha}=F_{(10,8),0.01}=\ 5.81.$$

Since calculated value of test statistic (= 3.8) is less than the critical value (= 5.81), that means calculated value of test statistic lies in rejection region as shown in Fig. 12.9, so we do not reject the null hypothesis and reject the alternative hypothesis i.e., we reject the claim at 1% level of significance. Thus, we conclude that samples provide us sufficient evidence against the claim so worker B is not more stable (or consistent) worker compared to A.

Now, you can try the following exercises.

2. Two sources of raw materials are under consideration by a bulb manufacturing company. Both sources seem to have similar characteristics but the company is not sure about their respective uniformity. A sample of 12 lots from source A yields a variance of 125 and a sample of 10 lots from source B yields a variance of 112. Is it likely that the variance of source A significantly differs to the variance of source B at significance level α = 0.01?
3. A laptop computer maker uses battery packs of two brands, A and B. While both brands have the same average battery life between charges (LBC), the computer maker seems to receive more complaints about shorter LBC than expected for battery packs of brand A. The computer maker suspects that this could be caused by higher variance in LBC for brand A. To check that, ten new battery packs from each brand are selected, installed on the same

models of laptops, and the laptops are allowed to run until the battery packs are completely discharged. The following are the observed LBCs in hours:

Brand A	3.2	3.7	3.1	3.3	2.5	2.2	3.2	3.1	3.2	4.3
Brand B	3.4	3.6	3.0	3.2	3.2	3.2	3.0	3.0	3.2	3.2

Assuming that the LBCs of both brands follows normal distribution, test the LBCs of brand A have a larger variance that those of brand B at 5% level of significance.

4. Two random samples drawn from two normal populations gave the following results:

Sample	Size	Mean	Sum of Squares of Deviation from the mean
Sample I	9	59	26
Sample II	11	60	32

Test whether both samples are from the same normal populations?